

Homework Flow & Curriculum Build — Cost Verification & Provider Reconciliation

Purpose. This document lets us *confirm* the costs the estimator shows: the per-session cost of the Homework Help (Tier 2) flow (Sections 1–7), and the one-off cost to build the whole KS3 + KS4 curriculum (Section 8). It checks every provider unit price against live provider documentation, then shows exactly how the biggest homework line — **TTS audio** — is counted on the **measured average** narration length, how image generation is costed at **one diagram per session** on the new **Gemini 3.1 Flash Image Light** model, and explains the o3 reasoning-token risk.

Bottom line up front: the unit prices are *correct and current*. Audio is billed on the **MEASURED AVERAGE** narration length (Section 2), route-specific — measured across real homework sessions rather than the old conservative 24,000-char maximum. Homework prices carry a **+15% buffer** and Inworld is priced on the **Developer plan (\$17 / 1M characters)**. Image generation is a single diagram per session on the new **Gemini 3.1 Flash Image Light (Nano-Banana-Light)** model at **\$0.05/image** (Section 3), down from \$0.10. The figure to keep validating on a real invoice is o3's **hidden reasoning tokens** (Section 4.3).

1. Provider unit prices — confirmed against documentation (July 2026)

Service	Model	Published list price	In estimator	Confirmed
Reasoning LLM	o3	\$2.00 / 1M input · \$8.00 / 1M output	\$2 / \$8	✓ matches OpenAI
Prose / vision LLM	gpt-4.1	\$2.00 / 1M input · \$8.00 / 1M output	\$2 / \$8	✓ matches OpenAI
Diagram image	Gemini 3.1 Flash Image Light (Nano-Banana-Light)	\$0.05 / image	\$0.05	✓ matches
Text-to-speech	Inworld TTS-2	\$17.00 / 1M characters (Developer plan)	\$17	✓ matches Inworld
Safety	omni-moderation	Free	\$0	✓

The +15% safety buffer (homework). Every homework price above is multiplied by **1.15** inside the estimator before any figure is shown, as an internal margin for retries, prompt drift and future price moves. So the effective rates the estimator charges against are: o3 / gpt-4.1 **\$2.30 / \$9.20** per 1M, Gemini **\$0.0575** per image, Inworld **\$19.55** per 1M characters. Throughout, we show **both** columns — "*list*" (the provider's published price, for reconciliation) and "+15%" (what the estimator displays). The one-off **curriculum build** (Section 8) is costed separately.

2. Text-to-speech (Inworld TTS-2) — billed on the MEASURED AVERAGE narration length

Inworld bills **per character** sent to synthesis. Earlier this document costed audio on a conservative **24,000-char maximum**. We have since **measured** the characters the homework flow actually sends to TTS across real sessions (`homework_tts_char_count.py` , 6 sessions), and the estimator now costs audio on that **measured average** — and, crucially, **route-specific**: prose *explanation* sessions narrate 2–4× longer than *calculation* sessions, so the blended figure follows the calculation / explanation mix.

Route (what the question needs)	Measured avg TTS chars / session	% of the 24,000 ceiling	n
Calculation → o3 (Maths / Physics / Chemistry sums)	4,758	20%	3
Explanation → gpt-4.1 (English, Biology prose)	12,611	53%	3
Blended average (50/50 default)	8,685	36%	6

Cost of the blended 8,685-char average: list $8,685 \div 1,000,000 \times \$17 = \$0.1476 \cdot +15\% = \0.1698 . Per route: calculation $4,758 \times \$17/1M = \0.0809 list / **\$0.0930** (+15%); explanation $12,611 \times \$17/1M = \0.2144 list / **\$0.2465** (+15%).

The 24,000-char maximum remains a safe worst-case bound, but the measured average is **~2.8× lower**, so it is the realistic figure for a blended forecast. Inworld also gets cheaper with commitment, so a real audio invoice can only come in lower than the Developer-plan rate:

Inworld TTS-2 tier	\$ / 1M characters	Cost of the 8,685-char avg session
On-demand	\$25.00	\$0.217
Developer (what we estimate on)	\$17.00	\$0.148
Higher commit	\$12.50	\$0.109
Enterprise	\$10.00 (as low as \$5)	\$0.087 (↓ \$0.043)

3. Image generation (Gemini 3.1 Flash Image Light) — one diagram per session

Homework Help generates **exactly one diagram image per session**, regardless of walkthrough depth or how the question arrived. This is homework-only behaviour: a single supporting diagram is produced for the session and reused across the walkthrough steps rather than re-generated at each phase.

`images / session = 1    (fixed – one diagram per session)`

Because it no longer scales with walkthrough depth, this line is flat and small. The homework flow has moved to the new **Gemini 3.1 Flash Image Light (Nano-Banana-Light)** model, so the estimator now prices the image at **\$0.05** — down from **\$0.10** on the previous model:

Images / session	Cost (list @ \$0.05)	Cost (+15%)
1 (fixed)	\$0.05	\$0.0575

This is a minor provider line in a session. At one image, generation is **\$0.05 list / \$0.0575 buffered**. If in future the flow needs more than one diagram per session, raise the image assumption in the estimator and this line scales linearly.

4. o3 reasoning model — tokens and cost

o3 is only used for **calculation / derivation** questions (Maths, Physics, Chemistry, computational Science). Prose/explanation questions run on gpt-4.1. The build pipeline is five LLM calls (steps A · parallel B · demo C · 3A explanation · answer key D); A→B→C can regenerate on a self-check failure, so they carry a retry factor (default **1.15×**).

4.1 Per-call token assumptions (o3 route)

Output tokens below are set to **include** o3's hidden reasoning tokens — see Section 4.3.

Call	Input tok	o3 output tok	Runs / session	o3 cost (list)
A · Solution steps	700	3,000	1.15	\$0.0292
B · Parallel question	800	2,500	1.15	\$0.0248
C · Parallel demo (worked)	900	3,000	1.15	\$0.0297
3A · Explanation	600	2,200	1.00	\$0.0188
D · Private answer key	900	3,000	1.00	\$0.0258
Validate attempt	700	1,500	3.00	\$0.0402
Follow-up Q&A	800	1,500	1.00	\$0.0136
o3 subtotal	~7,160	~20,975		\$0.1821

Plus the fixed gpt-4.1 intake/classify calls that run in **every** session (image extract + analyze + match): **\$0.0118**.

4.2 Per-session o3 cost (the math)

o3 input : 7,160 tok ÷ 1,000,000 × \$2 = \$0.0143

o3 output: 20,975 tok ÷ 1,000,000 × \$8 = \$0.1678

o3 total (list) = \$0.1821 → +15% = \$0.2094

4.3 Why o3 can look cheaper than the docs — hidden reasoning tokens

The unit price is right (\$2/\$8); the risk is the token count, not the rate. o3 bills *invisible* reasoning tokens at the **output** rate. OpenAI's documentation notes a single o3 answer can burn **5,000–20,000 internal reasoning tokens**. Our model already assumes ~21,000 o3 output tokens for a whole calculation session (near the high end), but if o3 reasons *harder*, cost scales. This table shows a full **calculation** session (o3 LLM + the fixed gpt-4.1 intake + the measured-average calc audio \$0.081 + the one-image \$0.05 line) as reasoning load rises:

o3 reasoning load	o3 output tok / session	Calc session (list)	Calc session (+15%)
As modelled	~21,000	\$0.325	\$0.374
1.5× harder	~31,500	\$0.409	\$0.470
2× harder	~42,000	\$0.493	\$0.567
3× harder	~63,000	\$0.661	\$0.760

The o3 output-token counts are **editable** in the estimator's *Advanced assumptions* — tune them to a sample of real billed sessions and the estimate tracks reality.

5. Full homework-session confirmation table

Per-session cost by service, with **audio at the measured average** (route-specific) and **one real-time image**. **Calculation** sessions use o3; **explanation** sessions use gpt-4.1; the **blended** row is the estimator default (50/50, image intake, audio on).

Session type	o3	gpt-4.1	Gemini image (×1)	Inworld audio (avg)	Total (list)	Total (+15%)
Calculation (o3)	\$0.182	\$0.012	\$0.050	\$0.081	\$0.325	\$0.374
Explanation (gpt-4.1)	—	\$0.070	\$0.050	\$0.214	\$0.334	\$0.384
Blended (default)	\$0.091	\$0.041	\$0.050	\$0.148	\$0.330	\$0.379

Every number is reproducible from Sections 2–4 with the published unit prices — that is the confirmation: **provider price × quantity = the estimator's figure**. Audio is now a measured line (~35–55% of a session depending on route), not a worst-case ceiling; the single image line is minor.

6. Scaling to volume (monthly / yearly)

Using the **blended buffered** cost of **\$0.379** per session, scaled by each preset's monthly session count (the *range* uses the calculation-only and explanation-only bounds, \$0.374 → \$0.384):

Preset	Students × questions	Sessions / month	Monthly (blended)	Yearly (blended)	Monthly range
Reset to defaults	200 × 10	2,000	\$758	\$9,096	\$748 – \$768
Light usage	300 × 20	6,000	\$2,274	\$27,288	\$2,244 – \$2,304
Heavy usage	600 × 40	24,000	\$9,096	\$109,152	\$8,976 – \$9,216

Monthly figures use the blended-default per-session cost. The two levers that move these most: the explanation / calculation mix (prose sessions narrate 2–4× more audio) and the o3 reasoning-token count (Section 4.3). The Inworld commitment tier lowers audio further (\$17 Developer → \$12.50 higher commit → \$10 enterprise).

7. Reconciliation summary (homework)

- Unit prices are correct and current** (verified July 2026) — o3 & gpt-4.1 at \$2/\$8 per 1M tokens, Gemini 3.1 Flash Image Light at \$0.05/image, Inworld TTS-2 at **\$17 per 1M characters (Developer plan)**.
- Audio is now a measured-average line, route-specific** — calculation ~4,758 chars, explanation ~12,611 chars, blended ~8,685 chars ≈ \$0.148 list / \$0.170 buffered — measured across 6 real sessions, replacing the old 24,000-char maximum (which remains a safe worst-case bound, ~2.8× higher).
- Image generation is a single diagram per session** — one **Gemini 3.1 Flash Image Light (Nano-Banana-Light)** image at **\$0.05 list / \$0.0575 buffered**, down from \$0.10, a minor line that no longer scales with walkthrough depth.
- Homework carries a +15% buffer**, so the estimate stays deliberately conservative.
- Validate two things on real usage:** (a) the **average narrated characters** per session vs. the measured 4,758 / 12,611 / 8,685 figures — the dominant line; and (b) o3's reasoning-token count (Section 4.3).

Recommended checks before committing to a budget

- Confirm the **average narrated characters** per session (per route) against a larger sample than 6.
- Pull the **actual o3 output-token count** from a dozen real calculation sessions vs. ~21,000/session.

- Decide the Inworld **commitment tier** (Developer \$17 → higher commit \$12.50 → enterprise \$10).

8. Full curriculum build — one-off content generation (separate from homework)

This is **not** a per-homework-session cost. It is the **one-off** cost to generate every lesson across the KS3 + KS4 curriculum, costed on **gpt-4.1** (the most expensive content model). Like homework, this build has now moved to the cheaper new image model for its ~30 whiteboard images per lesson.

8.1 Basis and the image-model change

This is the one-off cost to generate every lesson across KS3 + KS4, costed on **gpt-4.1** (the most expensive content model). Each lesson includes its script, narration audio, helpers, and **~30 whiteboard images**.

The only input that moved from the previous build is the **image model**: the ~30 images per lesson now run on the new **Gemini 3.1 Flash Image** model, dropping the image cost from **~\$1.89 to ~\$1.05 per lesson** (source: *General Cost Estimation Document (Updated).pdf*).

Image-model change, isolated. Nothing else in the build changed — same gpt-4.1 content model, same script/audio/helpers. The cheaper images alone lower the whole-curriculum total from **\$2,410.8 – \$2,881.2 to \$1,916.9 – \$2,290.9**; the per-subject figures are below.

8.2 Per key stage

Key Stage 3

Subject	Units	Lessons	Cost
Mathematics	6	57	\$278.7 – \$333.1
English	8	27	\$132.0 – \$157.8
Physics	7	43	\$210.3 – \$251.3
Chemistry	8	32	\$156.5 – \$187.0
Biology	6	38	\$185.8 – \$222.1
Total	35	197	\$963.3 – \$1,151.3

Key Stage 4

Subject	Units	Lessons	Cost
Mathematics	6	74	\$361.9 – \$432.5
English	11	38	\$185.8 – \$222.1
Physics	7	34	\$166.3 – \$198.7
Chemistry	10	49	\$239.6 – \$286.4
Biology	—	—	—
Total	34	195	\$953.5 – \$1,139.6

8.3 Whole curriculum

Whole curriculum (KS3 + KS4)	Units	Lessons	Cost
Combined	69	392	\$1,916.9 – \$2,290.9

Down from \$2,410.8 – \$2,881.2 on the previous image model. Biology KS4 was not in the source curriculum count, so it is left blank rather than estimated. The combined total is the sum of the two Key-Stage totals.

9. Live lesson session — why the LLM call count runs 16 to 24

The live lesson cost is not a flat per-minute rate. It is driven almost entirely by **how many LLM calls the check-in questions trigger**, so the call count is derived here rather than assumed.

9.1 Lesson shape

A lesson is **4 segments × 6 chunks = 24 chunks**. Chunks are narration; they do not each carry a question. `_apply_segment_checkin_rhythm` (`app/modules/content_generator.py`) marks exactly **two pause points per segment** — one mid-segment (chunk index 2, 3 or 4, chosen at random) and one on the last chunk — each asking **one question**.

	Per segment	Per lesson
Chunks	6	24
Check-in pause points	2	8
Questions asked	2	8

Worked examples and the quiz are excluded: they are served from cache and make no live call, and their text is not spoken live, so they contribute neither LLM tokens nor TTS characters.

9.2 The three calls behind one question

Each check-in is generated fresh at lesson time — nothing is read from the pre-built `QuestionBank`, which serves the practice quiz and the pre/post assessment instead. All three calls route through `POST /lessons/{id}/ask-question`.

Call	Fires	Capped?
1. Generate the question	Always — written against the chunk (or the whole segment, on the last chunk)	—
2. Spoken feedback	Always — the prompt branches on correct/incorrect, so a right answer still costs a call	—
3. Re-explain the concept	Only after a wrong answer	Max 1 per question ; the lesson then auto-advances

Floor = $8 \times 2 = 16$ calls. Ceiling = $8 \times 3 = 24$ calls.

9.3 Why 24 is reachable, and why nothing exceeds it

The floor of 16 assumes a student answers **all eight check-ins correctly, first time**. That is the best case, not the typical one. Four things push a real session toward the ceiling:

1. **Two-option multiple choice.** Check-ins offer exactly 2 options, so a student who is guessing is wrong about **half** the time. Blind guessing alone lands on ~4 re-explanations — **20 calls** — which makes 20, not 16, the statistically neutral midpoint.
2. **This tier assumes no prior knowledge.** The Foundation script is written for students starting from zero, which is precisely the cohort most likely to miss a check-in. The content tier and the wrong-answer rate are correlated, so this tier skews to the upper half of the range.
3. **“Explain again” is one tap and free to the student.** It is the only route to hearing the idea a second time, so when it is offered it is usually taken. Offered $\approx$ taken, in practice.
4. **Wrong answers cost more on both sides.** The re-explain is the largest prompt of the three (~4,700 characters in) and produces the longest spoken reply (~700 characters), so each one adds LLM *and* voice cost together.

The ceiling holds at 24 because the re-explain is capped at one per question in code — a second attempt is never offered, and the lesson advances as soon as the single re-explanation finishes speaking. The only way past 24 is the free-text answer path, which inserts an extra on-topic validation call; multiple choice is the default and free-text is not used for check-ins.

9.4 Cost across the range

Prompt sizes are measured from the actual prompt builders, including the backend system prompt and wrapper added to every call. Narration audio is pre-generated and cached, so the voice line covers only text composed live — the question, the feedback, and any re-explanation.

Scenario	Live calls	Tokens in	Tokens out	LLM (gpt-4.1)	Voice chars	Voice (Inworld)	Total / student
Every check-in correct	16	13,662	800	\$0.039	3,200	\$0.063	\$0.10
Half wrong — planning case	20	18,371	1,500	\$0.056	6,000	\$0.117	≈ \$0.17
Every check-in wrong	24	23,080	2,200	\$0.073	8,800	\$0.172	\$0.25

Budget on \$0.17 per student per lesson; size headroom for \$0.25.

This supersedes the previous flat estimate of ~25 calls / ~12,600 voice characters / \$0.41. That figure was a single measured session with no structural derivation, and it double-counted the pre-generated narration as a live voice cost.

Sources

- OpenAI o3 API pricing (\$2 / \$8 per 1M tokens; hidden reasoning tokens billed at output rate): <https://developers.openai.com/api/docs/pricing> · <https://pricepertoken.com/pricing-page/model/openai-o3>
- Inworld TTS-2 pricing (Developer plan \$17 / 1M characters; on-demand \$25, tiered to \$10 enterprise): <https://inworld.ai/pricing> · <https://www.buildmvpfast.com/api-costs/ai-voice>
- Measured TTS characters: `homework_tts_summary.pdf` / `homework_tts_char_count.py` (6 sessions, 2026-07-10).
- Curriculum build figures: `General Cost Estimation Document (Updated).pdf` (per-lesson gpt-4.1 with the new image model), reconciled against `General Cost Estimation Document.pdf` (previous image model).

Figures are grounded in the estimator's editable token / character / image assumptions; tune those to your real provider contracts and billed usage.